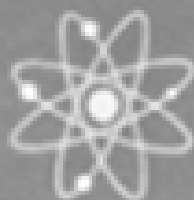


LRL Accelerators

THE 184-INCH SYNCHROCYCLOTRON

**LAWRENCE
RADIATION
LABORATORY**

UNIVERSITY OF CALIFORNIA, BERKELEY, CALIFORNIA



LRL Accelerators

THE 184-INCH SYNCHROCYCLOTRON

LAWRENCE
RADIATION
LABORATORY

UNIVERSITY OF CALIFORNIA, BERKELEY, CALIFORNIA

The Project Gutenberg eBook of LRL Accelerators, The 184-Inch Synchrocyclotron

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: LRL Accelerators, The 184-Inch Synchrocyclotron

Author: Lawrence Radiation Laboratory

Release date: August 10, 2010 [eBook #33397]

Language: English

Other information and formats: www.gutenberg.org/ebooks/33397

Credits: Produced by Erica Pfister-Altschul, Mark C. Orton, and the
Online Distributed Proofreading Team at <http://www.pgdp.net>

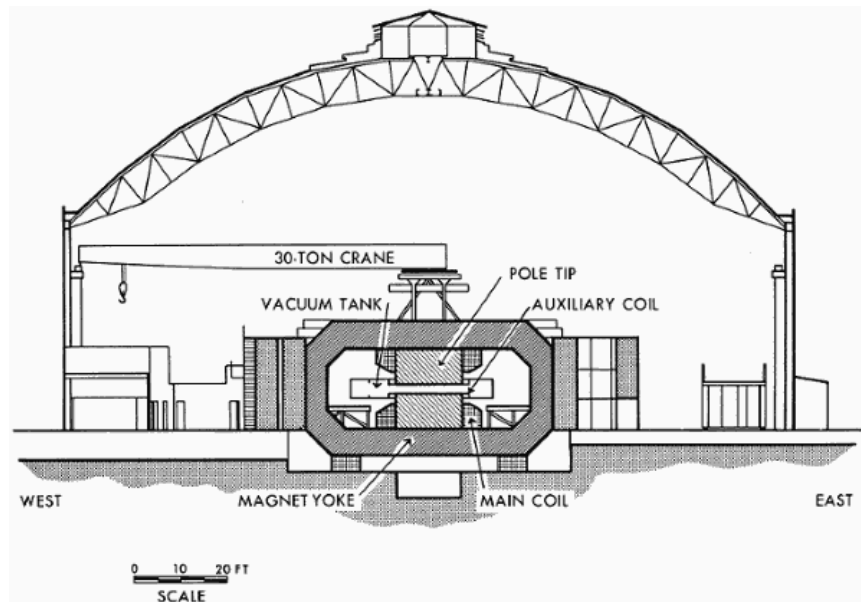
*** START OF THE PROJECT GUTENBERG EBOOK LRL ACCELERATORS, THE 184-INCH
SYNCHROCYCLOTRON ***

LRL Accelerators

THE 184-INCH SYNCHROCYCLOTRON

LAWRENCE RADIATION LABORATORY

UNIVERSITY OF CALIFORNIA, BERKELEY, CALIFORNIA



Synchrocyclotron Building

Contents

Page

[THE 184-INCH SYNCHROCYCLOTRON 2](#)

[PRINCIPLE OF OPERATION OF A CONVENTIONAL CYCLOTRON 3](#)

[THE PRINCIPLE OF PHASE STABILITY 6](#)

[DESIGN AND CONSTRUCTION OF THE 184-INCH SYNCHROCYCLOTRON 8](#)

[Magnet 8](#)

[Vacuum System 9](#)

[Ion Source 10](#)

[Radiofrequency System 10](#)

[Internal Targets and Beam Extractor 12](#)

[CYCLOTRON EXPERIMENTS 15](#)

[Nuclear Physics 15](#)

[Biophysics 18](#)

[Nuclear Chemistry 19](#)

[BIBLIOGRAPHY 20](#)

THE 184-INCH SYNCHROCYCLOTRON

His success with the 60-inch cyclotron in 1939 led Dr. E. O. Lawrence to propose a much more powerful accelerator, one which could produce new types of nuclear rearrangements and even create particles. Grants totaling \$1,225,000 permitted work to start on the 184-inch cyclotron in August 1940.^[1] It was designed to accelerate atomic particles to an energy of 100 million electron volts (Mev), five times that possible with the 60-inch machine.

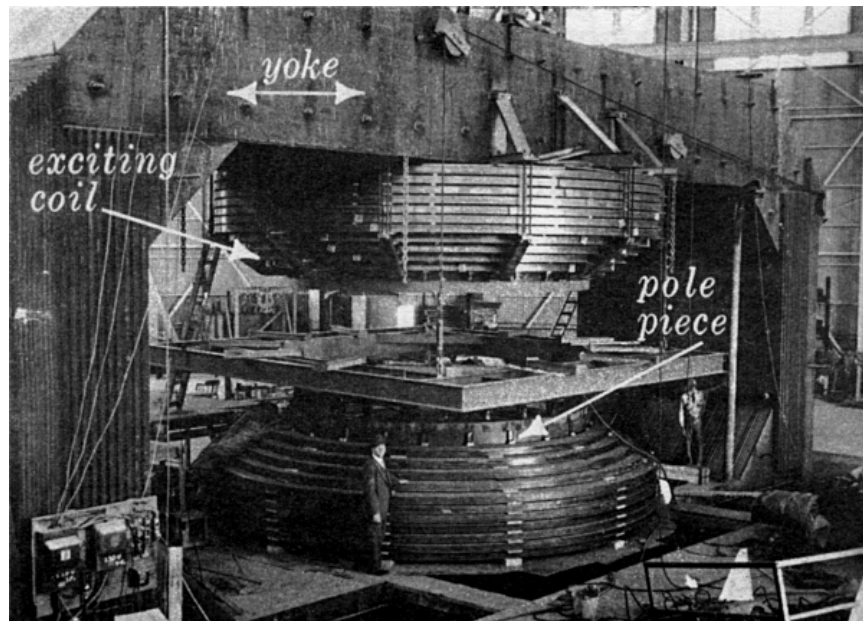


Fig. 1. The electromagnet under construction during the period 1940 to 1942.

Before the new cyclotron could be finished World War II began. Construction on the cyclotron was therefore halted. However, because of interest in separating the isotopes of uranium by the electromagnetic method, work on the giant magnet continued at an even faster pace. This magnet would contain 3700 tons of steel in its yoke and pole pieces, and 300 tons of copper in its exciting coils (Fig. 1). By May 1942 the magnet was completed. During that summer it was used in a pilot plant to separate the first significant amounts of U^{235} ever obtained. The 184-inch magnet remained in use in a research and development program at Berkeley until the end of the war, supplying information to Oak Ridge, Tennessee, where a large separation plant had been erected.

Construction on the rest of the cyclotron was resumed in 1945. By that time a new principle had been discovered which made it possible to obtain ion beams of much higher energy than originally hoped for. Yet a considerably lower accelerating voltage could be used. This important discovery was made independently by Dr. V. Veksler in Russia and by Dr. Edwin M. McMillan, present Director of the Lawrence Radiation Laboratory. Before attempting to discuss this principle, we should first review the operation of a conventional cyclotron.

PRINCIPLE OF OPERATION OF A CONVENTIONAL CYCLOTRON

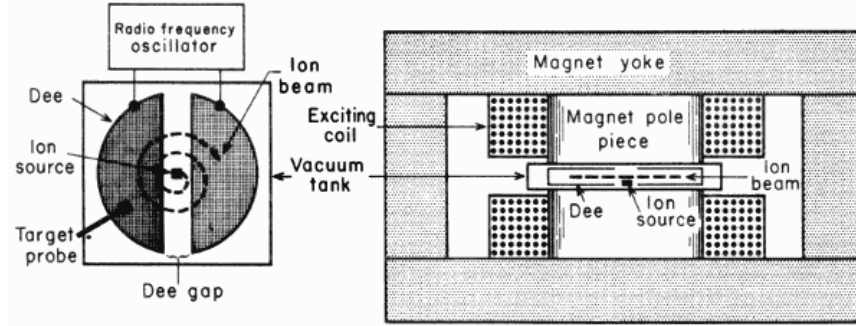


Fig. 2. Basic parts of a cyclotron.

The main parts of a cyclotron are represented in Fig. 2. Charged particles (ions) are accelerated inside an evacuated tank. This is to prevent the beam from colliding with air molecules and being scattered. The vacuum tank is placed between the poles of an electromagnet, whose field bends the ion beam into a circular orbit.

The operation begins when the ions are introduced into the region between two accelerating electrodes, or "dees."^[2] Because the ions carry a positive electric charge, they are attracted toward that dee which is electrically negative at the moment. Were it not for the magnetic field, the ions would be accelerated in a straight line; instead they are deflected into a circular path back toward the dee gap. By the time the ions again reach the dee gap, the sign of the electric potential on the dees is reversed, so that now the ions are attracted toward the opposite dee.

As this process of alternating the electric potential is repeated, the ions gain speed and energy with each revolution. This causes them to spiral outward. Finally they strike a target inserted into their path or are extracted from the cyclotron for use as an external beam.

The time required for an ion to complete one loop remains constant as it spirals outward. This is because its velocity increases sufficiently to make up for the increased distance it travels during each turn. This means that the electric potential applied to the dees must alternate at a constant frequency, called the "resonant frequency."

The resonant frequency f is given by the relationship

$$f = \frac{He}{2\pi mc} \quad (1)$$

where H , e , π , c , and m are constants. H is the strength of the magnetic field of the cyclotron, e is the electric charge carried by the ion, π equals 3.14, c is a conversion factor, and m is the mass of the ion. For example, the resonant frequency for protons accelerated in a 15,000-gauss magnetic field is 23.7 megacycles (Mc).^[3] We call such a rapidly alternating potential a "radiofrequency voltage" and the electronic circuit for producing it a "radiofrequency oscillator."

The energy E of an ion emerging from the cyclotron is given by

$$E = \frac{H^2 R^2 e^2}{2 mc^2} \quad (2)$$

where H , e , and m are as defined above, and R is the radius at which the beam is extracted. From this equation we see that for a given type of ion (where e and m are constant), the energy depends on the

diameter and strength of the magnet, but not directly upon the voltage applied to the dees.

The number of revolutions that an ion can make in a conventional cyclotron is limited to about 70 to 100. This is due to a very curious effect: as an ion is accelerated, its mass increases! [This phenomenon is explained by Einstein's special theory of relativity (see Fig. 3).] Referring back to Eq. (1), we see that if the ion mass (m) does not remain constant, but rather increases, then the resonant frequency (f) decreases. But since the dee potential continues alternating at a constant frequency, an ion soon begins to arrive "late" at the dee gap. By the time it has made about 70 to 100 turns an ion is so badly out of phase that it is no longer accelerated.

Suppose now that we want to obtain an energy of 10 Mev. Because an ion can make a maximum of about 100 turns, the accelerating potential would have to be about 100,000 volts. However, Professor Lawrence hoped to reach 100 Mev with the new 184-inch cyclotron. This meant that the accelerating voltage would have to be about 1,000,000 volts. Preventing such a high voltage from sparking promised to be one of many formidable engineering problems.

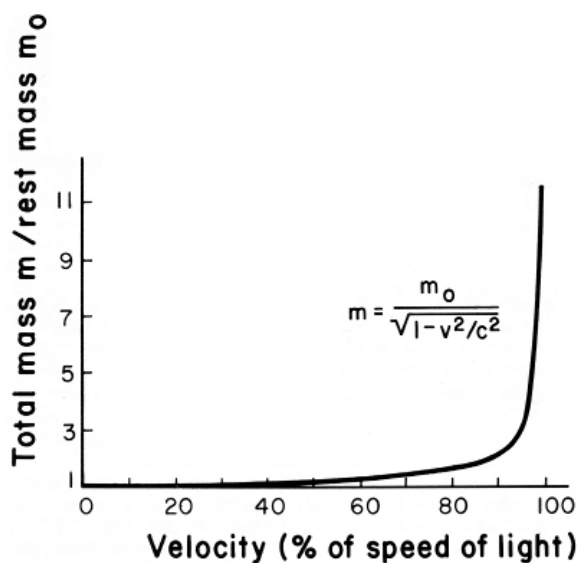


Fig. 3. Graph showing how the mass of an object increases as its velocity approaches that of light.

THE PRINCIPLE OF PHASE STABILITY

Fortunately, Drs. Veksler and McMillan showed that relatively low dee voltages can be used to accelerate ions to very high energies. This is possible if the oscillator frequency is continuously decreased to keep it in synchronism with the decreasing rotational frequency of the ions. This would allow an ion to make many revolutions without becoming out of phase. This principle of phase stability was experimentally verified with the 37-inch cyclotron before being incorporated into the design of the 184-inch machine. Because it utilizes this principle, this machine has usually been referred to as a "synchrocyclotron" or "frequency-modulated cyclotron." However, it is sometimes called simply a "cyclotron."

The 184-inch synchrocyclotron was first operated in November 1946. With a maximum dee voltage of only 20,000 volts, it accelerated deuterons to 190 Mev and alpha particles to 380 Mev.^[4] In 1949 it was modified to permit production of 350-Mev protons also.

Between 1955 and 1957 the synchrocyclotron was rebuilt so that now the following energies can be obtained:

<u>Protons</u>	<u>Deuterons</u>	<u>Alpha Particles</u>	<u>Helium-3 nuclei</u> ^[5]
730 Mev	460 Mev	910 Mev	1140 Mev

In reaching an energy of 730 Mev a proton, for example, makes 75,000 revolutions in just 6 milliseconds (msec). It travels a distance of 450 miles and attains a velocity of 152,000 miles per second, or 82% of the speed of light! During this brief journey its mass increases 75%, giving very convincing evidence for the validity of Einstein's theory. Similar data for other ions may be found in the appendix.

DESIGN AND CONSTRUCTION OF THE 184-INCH SYNCHROCYCLOTRON

Magnet

During the rebuilding of the cyclotron, the diameter of the magnet pole pieces was increased from 184 to 188-3/4 inches. Also, the pole gap at the center was reduced from 21 to 14 inches. These changes increased the weight of steel in the magnet from 3700 to 4000 tons.

The main exciting coils, which contain 1300 turns of copper-bar conductor each, were not altered. Two auxiliary coils containing 425 turns each were added. This brought the total weight of copper from 300 to 340 tons. The coils are layer-wound around the pole pieces close to the pole gap. Other data about the coils are given in the appendix.

The effect of these modifications was to increase the field strength at the center of the pole gap from 15,000 to 23,400 gauss. This increase made it possible to obtain the higher-energy ions.

Power is supplied to the coils by two motor generator sets, which produce the direct current required for a steady magnetic field. The direct current from the motor generators is regulated so that the magnetic-field fluctuation is less than one part in 10,000. This is necessary if one wants an external beam of nearly uniform energy.

In order to prevent the beam from becoming unstable and striking the dee, the magnetic field must be strongest at the center and decrease radially (Fig. 4a). With flat pole faces the field does not decrease uniformly. To give the desired rate of decrease, the pole faces are shimmed with concentric steel rings of varying thickness, as shown in Fig. 4b. In a radially decreasing magnetic field, the lines of magnetic flux bow outward, as represented in Fig. 4b. Ions moving in a magnetic field are deflected at right angles to these flux lines. Ions above the midplane of the cyclotron are directed downward; those below the midplane are directed upward. In this way an ion oscillates about the midplane and vertical focusing is achieved.

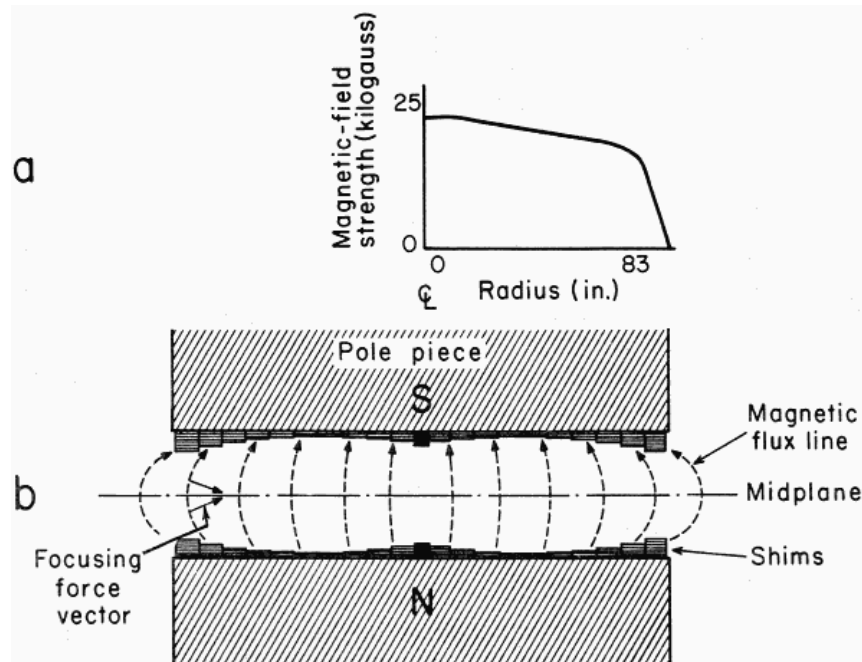


Fig. 4. (a) Plot of magnetic-field strength vs radius. The field strength decreases gradually out to a radius of about 83-in., after which it falls off sharply. This point marks the maximum usable radius for particle orbits. Further out they are unstable. (b) Magnetic flux lines are represented as broken arrows, and focusing forces as solid arrows. An ion above the midplane is directed downward, while an ion below the midplane is directed upward.

Radial focusing is accomplished in a somewhat analogous manner. If the magnetic field decreases with radius, radial restoring forces are established. An ion at too large a radius is directed inward, and an ion at too small a radius is directed outward. In this fashion, the ion oscillates about the synchronous orbit. Thus, radial focusing is achieved.

Vacuum System

The vacuum tank (acceleration chamber) is a steel box 20 × 25 ft and 4 ft high. It is evacuated to a pressure of 10^{-5} millimeter of mercury (about one 100-millionth of atmospheric pressure). The pumping equipment consists of six oil-diffusion pumps and four mechanical vacuum pumps. The pumping speed of the six 20-in. oil-diffusion pumps is a total of 20,000 liters/sec.

Ion Source

The ion source is a simple arc-type. Hydrogen gas is allowed to leak into the ion-source enclosure near a tungsten filament, which is heated to incandescence. Electrons emitted by the filament knock off electrons from hydrogen atoms, leaving free protons. The protons then escape into the acceleration chamber through a hole in the ion-source housing. Once inside, the protons are accelerated by the dee potential.

Deuterons or alpha particles are obtained in a similar fashion using deuterium or helium gas in place of hydrogen.

Radiofrequency System

The 184-inch synchrocyclotron has a single dee instead of the double-dee arrangement described above for illustrative purposes. The accelerating electric field is developed between the dee and a dummy dee which is grounded to the vacuum tank. Using a single dee does not change the principle of operation, yet it offers the advantage of allowing more space for auxiliary equipment inside the vacuum tank. Also, the construction is much simpler. The dummy dee is not essential for operation, but it does improve performance.

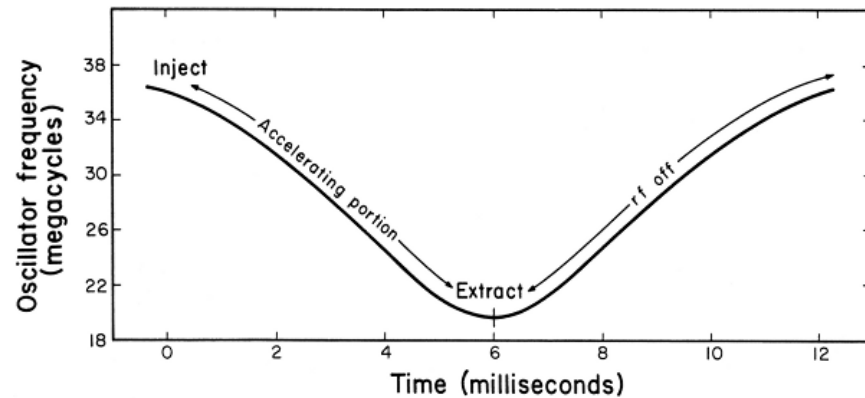


Fig. 5. Radiofrequency cycle for accelerating protons. Sixty-four such cycles are repeated each second.

Radiofrequency power is supplied to the dee by a vacuum-tube oscillator. The frequency of oscillation must decrease during the acceleration cycle, as indicated above. For protons, the frequency at the start of acceleration is 36 megacycles (Mc). At the end of acceleration the frequency is only 18 Mc (see Fig. 5). This change in frequency is achieved by varying the electrical capacitance in the tuned circuit of the oscillator. (This is what you do when you dial a different station on a radio.) This tuned circuit, which is called the cyclotron resonator, is shown in Fig. 6.

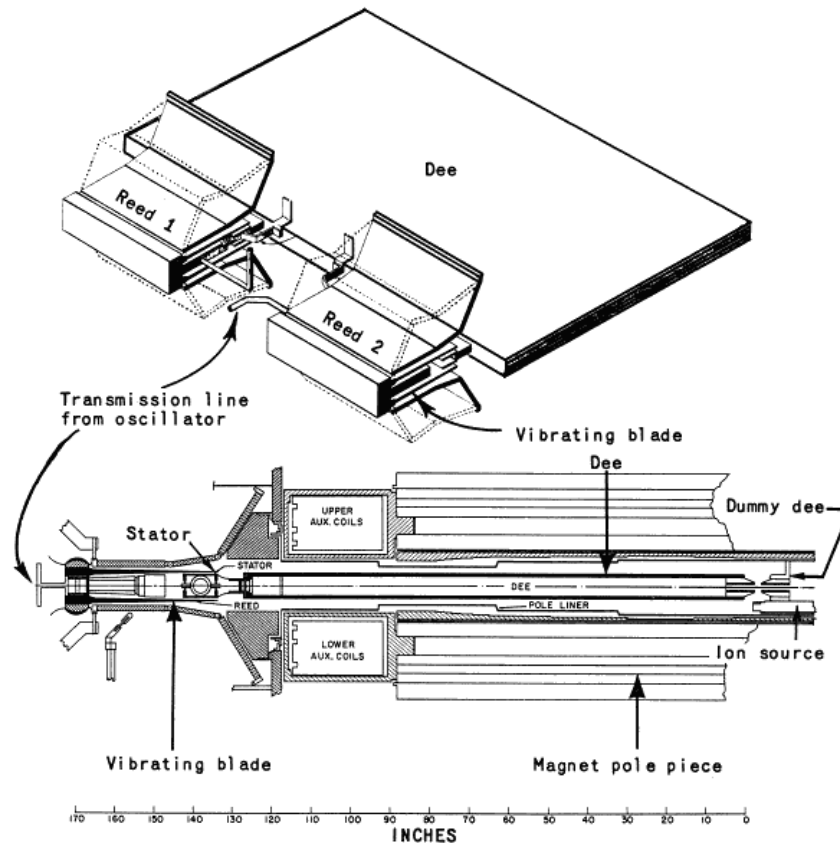


Fig. 6. Cyclotron resonator.

Because the frequency must change over such a wide range (from 36 to 18 Mc), the electrical capacitance must be varied by a factor of 20 to 1. This is done by a variable capacitor of unique design. It resembles two giant tuning forks. As the blades of the tuning forks vibrate, the capacitance is alternately increased and decreased by the required amount.

These two tuning forks must be kept in step with great precision. This is to prevent the oscillator from exciting lateral rf resonances. With a cyclotron of this size, this is a problem. These resonances, if excited, would cause loss of beam. The method for keeping the blades moving together is as follows: The blades are made to vibrate at their resonant frequency, which is approximately 64 cycles per second. One set of blades operates at its natural frequency as a tuning-fork oscillator. The second set of blades is driven from an amplified sample of the signal from the first; its natural period is adjusted automatically to equal that of the first. The amplitude of each set is regulated to within 0.003 in.; the phase angle between the blades is regulated to within 1 deg.

Ions are accelerated only when the radiofrequency is decreasing (Fig. 5). The remaining portion of the cycle is "dead time." Thus, 64 pulses, each of about 500 microseconds' duration, are obtained every second. The average ion current of a pulsed beam is much less than for a continuous beam, such as that obtained from a conventional cyclotron (see Table I). This is part of the price paid for higher energies.

Internal Targets and Beam Extractor

The simplest target is one placed inside the vacuum tank where the circulating beam will strike it. The target may be any substance that the physicist or chemist wants to irradiate. The target material is attached to a movable probe. If the experimenter wants to use the full-energy beam, he places the target at the

maximum usable radius of the circulating beam (82 inches). However, if he desires to use ions having less than the maximum energy, he inserts the target further into the cyclotron so that it is intercepted sooner.

TABLE I

Comparison of external-beam energy and current for a synchrocyclotron and a conventional cyclotron			
184-Inch Synchrocyclotron			
	Protons	Deuterons	Alpha particles
Beam energy — maximum (Mev)	730	460	910
Beam intensity — peak current (μa) ^[a]	120	120	40
Beam intensity — average current (μa)	0.75	0.75	0.25
60-Inch Cyclotron			
Beam energy — maximum (Mev)	12	24	48
Beam intensity — peak current (μa)	100	150	100
Beam intensity — average current (μa)	70	80	60

[a] μa = microampere

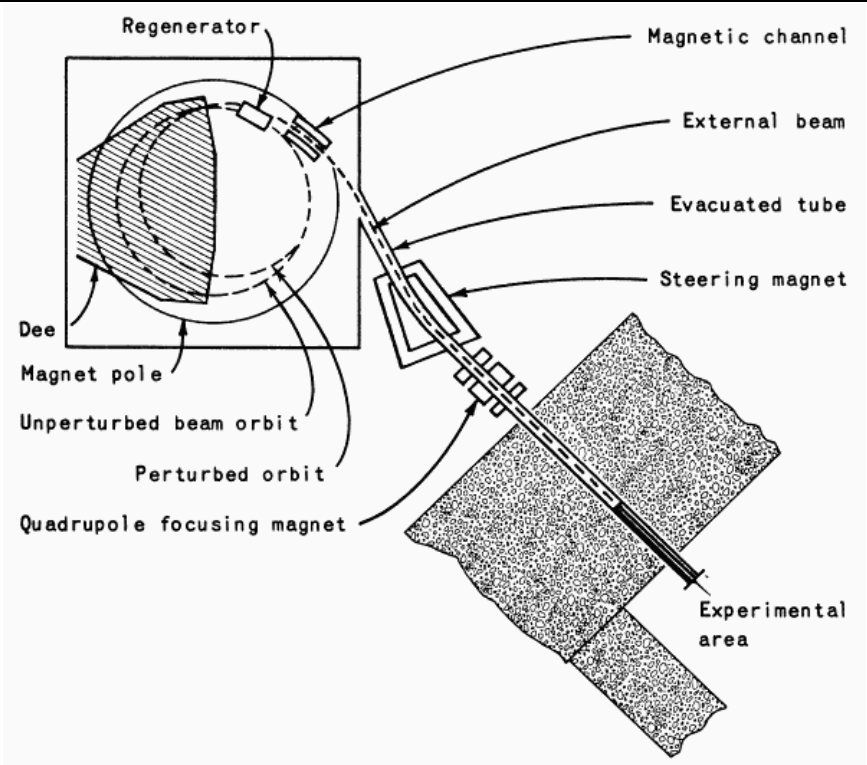


Fig. 7. Plan view of the cyclotron, showing the method for obtaining an external beam of protons, deuterons, or alpha particles.

Some experiments require an external beam of protons, deuterons, or alpha particles. A beam of this type can be brought out of the machine by means of a LeCouteur regenerator (Fig. 7). The construction of the regenerator is very simple. It is made of a number of steel laminations of various sizes. What the regenerator does is perturb the magnetic field of the cyclotron at one radial position. Each time the beam passes through the regenerator it receives a kick. With each kick the beam builds up its radial amplitude, until finally it enters a magnetic channel. This channel focuses the beam and steers it outside the main magnetic field. Once outside, the beam travels through an evacuated tube, which is integral with the main vacuum tank. By means of a steering magnet, the beam can be sent into either the physics cave or the medical cave. (These experimental areas are called "caves" because they are rooms inside the massive concrete shielding wall.)

Other experiments may require an external beam of mesons.^[6] A meson beam is obtained in the following way (Fig. 8): A movable target such as a block of carbon is placed inside the cyclotron near the end of the outward-spiraling proton beam. When the proton beam hits this target, a shower of mesons is produced. These mesons are bent in various directions by the main magnetic field. Some of them pass through a thin metal window in the vacuum-tank wall and are focused by a magnetic lens into a beam. This meson beam then travels through a hole in the concrete shielding wall into the meson cave. The maximum intensity of this extracted meson beam depends on both the charge and energy desired. Beams of more than 100,000 mesons per second have been obtained through an aperture 4×4 in. in the shielding wall.

CYCLOTRON EXPERIMENTS

Nuclear Physics

About 86% of the operating time of the 184-inch synchrocyclotron is devoted to experiments in nuclear physics. Most of the experiments study the production and interaction of π mesons. These particles are considered to be essential factors in the intense but short-range forces that bind the nucleus together. The three types of π mesons are designated according to their electric charge as π^+ , π^0 , and π^- .^[7] These mesons materialize only in high-energy nuclear collisions.

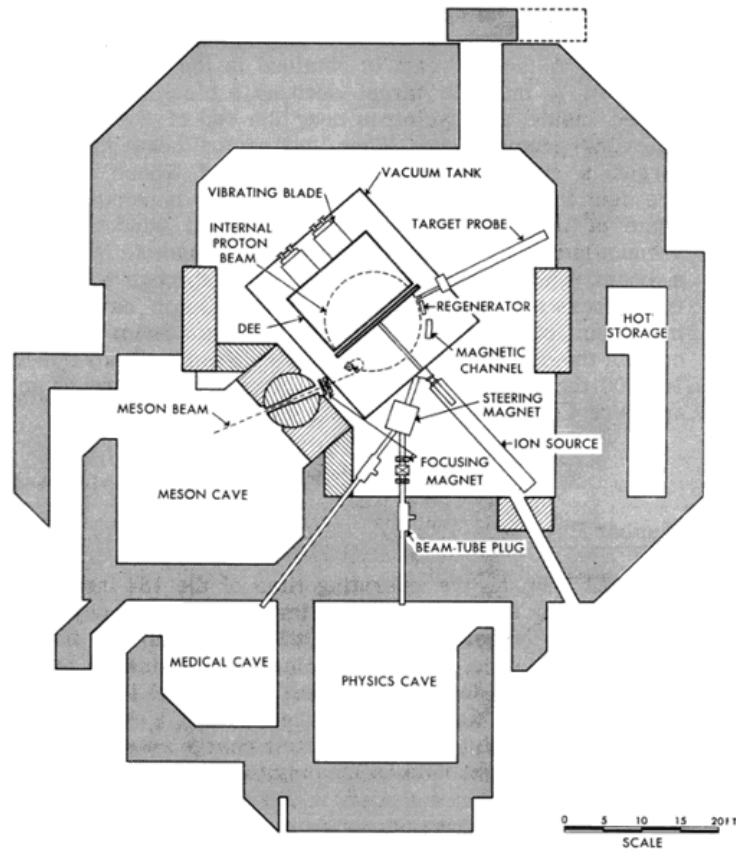


Fig. 8. Method for obtaining external meson beam.

Of great importance are those experiments that determine the probability of producing each of the three types of mesons in a nuclear collision. This type of experiment is repeated for different beam energies and target elements. Other experiments measure the energy and direction of emission of π mesons from a target.

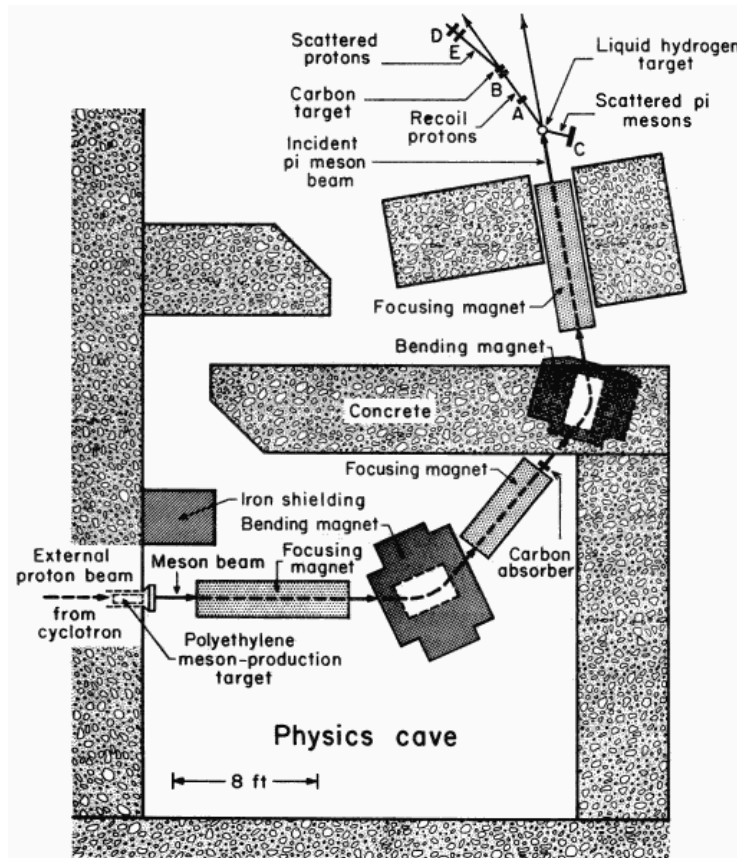


Fig. 9. A typical experiment. Scintillation counters at A, B, C, D, and E record the passage of charged particles.

A typical π -meson experiment is represented in Fig. 9. The purpose of this experiment was to detect the spin directions of protons as they are knocked out of a liquid hydrogen target by a π -meson beam. (Like the earth, a proton spins on its axis.) An extracted proton beam from the cyclotron enters the physics cave from the left, striking a polyethylene target and producing π mesons. A beam of these mesons is formed by a series of two bending magnets and three focusing magnets. This beam passes through a carbon absorber to remove unwanted particles. The meson beam then strikes the liquid hydrogen target. A few of the incoming mesons scatter, knocking protons out of the liquid hydrogen. Scintillation counters at A and B record the passage of a proton, thus defining its direction. The scattered mesons are counted by a scintillation counter at C. A few of the protons scatter off the carbon target and are detected by counters at E and D. From the detection of such events, the spin directions (polarization) of the recoil protons can be analyzed. In this way, more is learned about the fundamental π -proton interaction.

Further studies of the interactions of π mesons are made in the meson cave. Other experiments performed there are concerned with μ mesons. The μ meson (muon) is a particle created in the decay of a π meson and is the principal constituent of cosmic rays striking the surface of the earth. The muon is unstable, eventually undergoing a radioactive decay into an electron. Although the muon does not experience nuclear forces, it can interact weakly with nuclei. The behavior of the muon is well understood, but its role as one of the elementary particles is unknown. That is, if the muon did not exist, what effect would this have on the structure of matter? The answer to this question, among others, is being sought by physicists using the 184-inch cyclotron.

Biophysics

Experiments in biophysics are conducted in the medical cave. In these the interest lies not in nuclear interactions but in the effect of ionizing radiation on living tissue. High-energy beams of particles can be used for selective destruction of specific areas of the brain. This permits physiological mapping of the functions of the brain in experimental animals. It further offers a therapeutic approach to the treatment of brain tumors. One of the important investigational programs is concerned with the relationship of the pituitary gland to the growth rate of certain cancers and to some endocrine disorders.

Nuclear Chemistry

For techniques of radiochemistry to be employed successfully, high interaction rates (and therefore high beam intensities) are needed. For this reason, chemistry targets are usually inserted right into the cyclotron so that they can be bombarded directly by the circulating beam. After the bombardment is completed the target is removed from the cyclotron. It is then taken to a chemistry laboratory and subjected to detailed chemical procedures. Individual elements are removed, and the radioactive isotopes of each element are identified by quantitative counting techniques. In some cases a mass spectrometer is used to analyze the products. Many deductions about the nature of the breakup of the target nucleus can be drawn from the pattern of the observed radioactive products. Sometimes the nucleus splits almost in half. This is called fission. More frequently smaller parts of the nucleus are split off. Two general types of reactions, known as spallation and fragmentation, are distinguished. One of the goals of this research is to learn more about the constitution of the nucleus and of the forces which bind the particles in the interior of the nucleus.

FOOTNOTES:

[1] The grants were as follows: Rockefeller Foundation—\$1,150,000; John and Mary Markle Foundation—\$25,000; The Research Corporation—\$50,000. The University of California added a guarantee of \$175,000 to bring the total building fund to \$1,400,000.

[2] In the first cyclotrons the electrodes were shaped like the letter "D."

[3] We have the values $H = 15,000$ gauss, $e = 4.8 \times 10^{-10}$ electrostatic units, and $m = 1.6 \times 10^{-24}$ gram. To find f , we write

$$f = \frac{15,000 (4.8) 10^{-10}}{2 (3.14)(1.6) 10^{-24} (3) 10^{10}},$$

$$f = 23.7 \text{ Mc.}$$

[4] A deuteron is the nucleus of an atom of heavy hydrogen and contains one proton and one neutron; it carries a single positive electric charge. An alpha particle is the nucleus of a helium atom and is made up of two protons and two neutrons; it carries two positive charges.

[5] The machine is equipped for helium-3 operation, but to date it has not been used for that purpose.

[6] Mesons are elementary particles intermediate in mass between the electron and proton.

[7] It may be interesting to note that the π^0 meson was discovered with this cyclotron in 1950. This was the first particle to be discovered with an accelerator. All particles that had been previously discovered were observed first in cosmic rays or some other form of natural radiation.

BIBLIOGRAPHY

1. Gerald A. Behman, Particle Accelerators: I. Bibliography, II. List of Accelerator Installations, UCRL-8050, January 1, 1958.
2. Samuel Glasstone, The Acceleration of Charged Particles, in *Sourcebook on Atomic Energy*, Second Edition (Van Nostrand, Princeton, 1958), Ch. IX.
3. M. S. Livingston, *High-Energy Accelerators* (Interscience Publishers, New York, 1954).
4. M. Stanley Livingston and Edwin M. McMillan, History of the Cyclotron, *Physics Today* 12, 18-34 (October 1959).
5. E. M. McMillan, Particle Accelerators, in *Experimental Nuclear Physics*, Emilio Segrè, Editor, Vol. III (Wiley, New York, 1959), Part XIII.
6. Bob H. Smith *et al.*, The Electrical Aspects of the UCRL 740-Mev Synchrocyclotron, UCRL-3779 Rev., October 2, 1957.
7. Robert L. Thornton, Frequency-Modulation and Radiofrequency System for the Modified Berkeley Cyclotron, UCRL-3362, April 3, 1956.
8. Robert R. Wilson, Particle Accelerators, *Scientific American* 198, 64-76 (March 1958).

APPENDIX

SUMMARY OF SPECIFICATIONS

<u>Present fields of research</u>	<u>% of time</u>
Nuclear physics	86
Nuclear chemistry	2
Biophysics	12
<u>Scheduled operation</u>	156 hours/week

Performance

Internal Beams

	<u>Protons</u>	<u>Deuterons</u>	<u>Alpha particles</u>	<u>Helium-3 ions</u>
Maximum energy (Mev)	730	460	910	1140
Energy spread (Mev)	55			
Beam intensity				
Average current (μ a)	0.75	0.75	0.25	
Peak current (μ a)	120	120	40	
Beam radius, maximum (in.)	82	82	82	82
Time required for acceleration (msec)	6	4.5	4.5	
Number of revolutions during acceleration	75,000	60,000	60,000	
Distance traveled during acceleration (miles)	450	360	360	
Velocity at maximum energy (% of speed of light)	82	60	60	69
Mass increase at maximum energy (% of rest mass)	75	25	25	40
Range of full-energy particles (in. of aluminum)	37	12	7	

External Beams

	Physics cave			Meson cave			
	Protons	Neutrons	π^+	π^+		π^-	
Energy (Mev)	730		310	100	250	100	300
Energy spread (Mev)	14			10	20	10	30
Beam area (cm ²)	25		40	100	100	100	100
Flux (particles/cm ² -sec)	2×10^{10}	5×10^5	5×10^4	1000	100	1500	1000

Acceleration chamber (vacuum tank)

Size

length (ft)	25
width (ft)	20
height (ft)	4

Material: mild steel

Operating pressure (mm Hg) 10^{-5}

Vacuum pumps: six 20-in. oil-diffusion pumps with 8-in. boosters:
one Beach-Russ 750-cfm; one Kinney 300-cfm; two Kinney 105-cfm.

Pumping speed of oil-diffusion pumps (liters/sec) 20,000

Magnet

Core diameter (in.) 184

Pole-tip diameter (in.) 188.75

Pole gap at center (in.) 14

Magnetic field strength (gauss)

at center 23,400

at radius of 82.2 in., where $n = 0.2$ 22,275

Weight of steel (tons) 4,000

	Main coils	Auxiliary coils
Magnet coils		
material	solid copper ($1/4 \times 4$ in.)	hollow copper ($1-3/16 \times 1-1/16$ in.)
weight of copper (tons)	300	40
number of turns (total)	2,600	425
ampere turns	1.9×10^6	1.1×10^6
current (amp)	1600	2800
voltage (v)	550	560
power (kw)	900	1600
coolant	oil	water

Radiofrequency system

Dee system

number of dees 1

size

length (in.) 126

width (in.) 180
height (in.) 48
material: 1/64-in.-thick copper, stretched over a stainless steel frame
dee aperture (in.) 4-3/16

Oscillator

type: self-excited grounded-grid 10
tube: one Machlett ML5681
dc input, operating condition (kw) 10
dee bias, maximum dc (v) 2000

	Protons	Deuterons	Alpha particles
rf duty cycle (%)	38	28	28
dee-to-ground voltage, peak (kv)	9	6	6

Frequency-modulation system

type: vibrating-reed (tuning-fork) capacitor number of units: two (two blades each)
blades
size
width (in.) 45
length (in.) 32
thickness: tapered from 1.4 to 0.06 in.
weight (lb) 500
vibrational frequency (cps) 64
electrical capacitance ($\mu\mu\text{f}$) 300 to 6,500
peak-to-peak excursion (in.) 1
minimum separation of blade and stator (mils) 50

	Protons	Deuterons	Alpha particles
frequency sweep (Mc)	36-18	18-13.5	18-13.5

Ion source: conventional open-arc type

Beam extractor

LeCouteur-type regenerator combined with magnetic channel

Building and facilities

Room dimensions
diameter (ft) 160
height (ft) 90
Crane
type: radial
capacity (tons) 30
overhead span (ft) 77
Concrete shielding: 15 ft thick on sides, 4 ft on top

History.

Design started: January 1940.

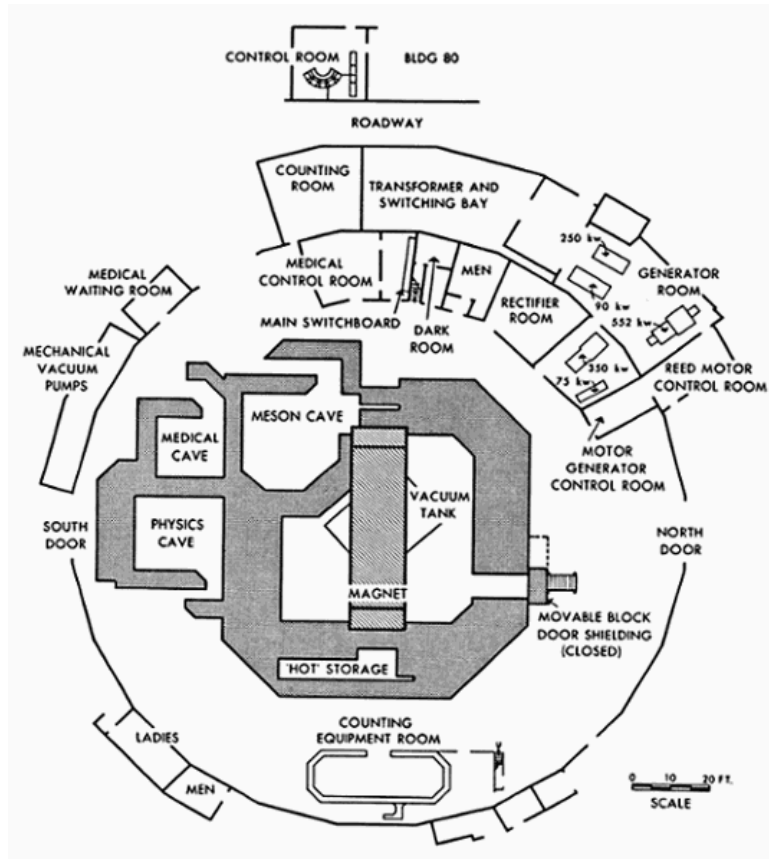
Construction started: August 1940.

First operation

for deuterons and alpha particles: November 1946.

for protons: December 1948.

Rebuilt: 1955-1957.

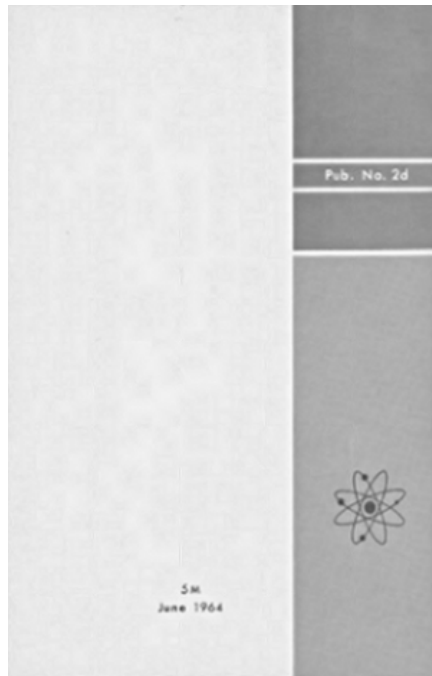


Synchrocyclotron Building

Pub. No. 2d

5M

June 1964



Transcriber's Note

Text footnotes have been moved to the end of the chapters, before the bibliography and appendix.

The following changes have been made to the printed text:

[Page 15](#): added closing quote (*are called "caves" because*)

[Page 19](#): *iostopes* corrected to *isotopes*

*** END OF THE PROJECT GUTENBERG EBOOK LRL
ACCELERATORS, THE 184-INCH SYNCHROCYCLOTRON ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

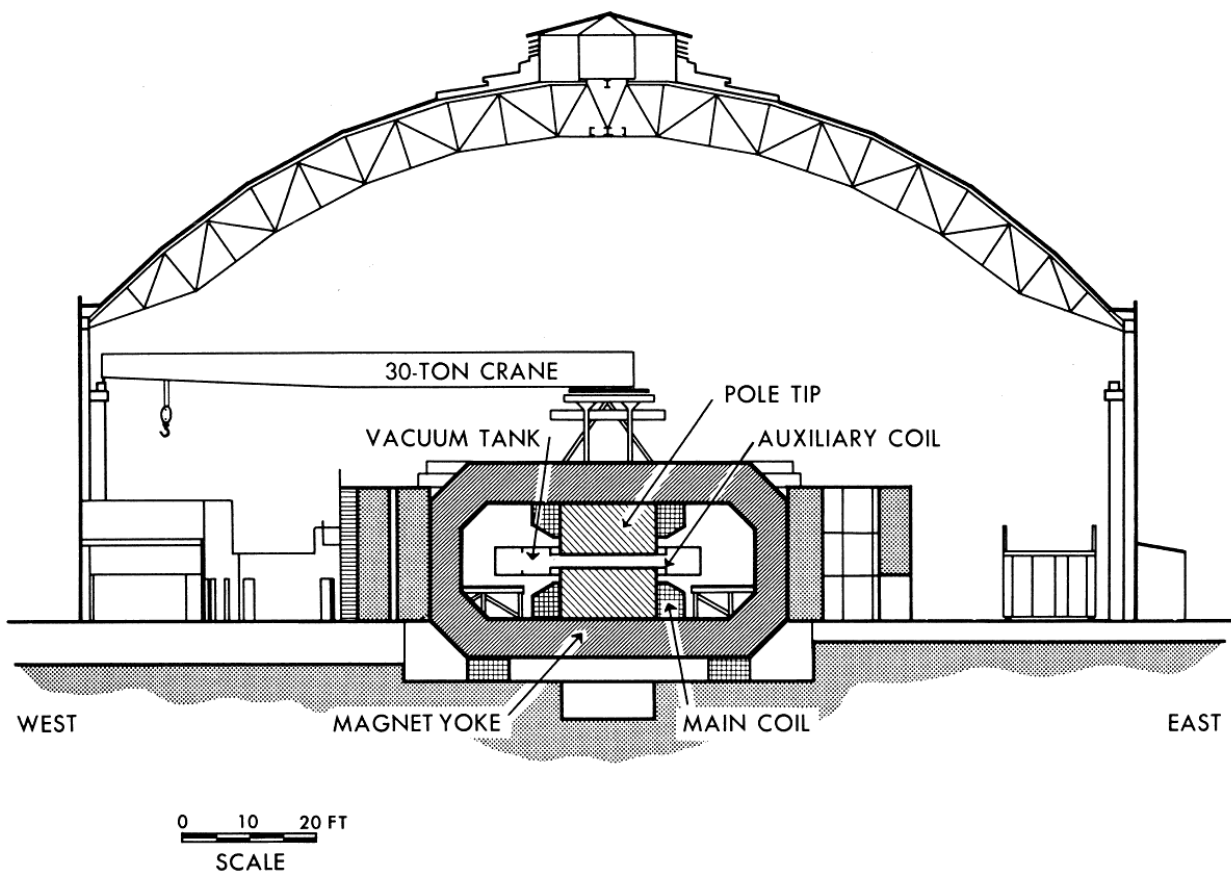
Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

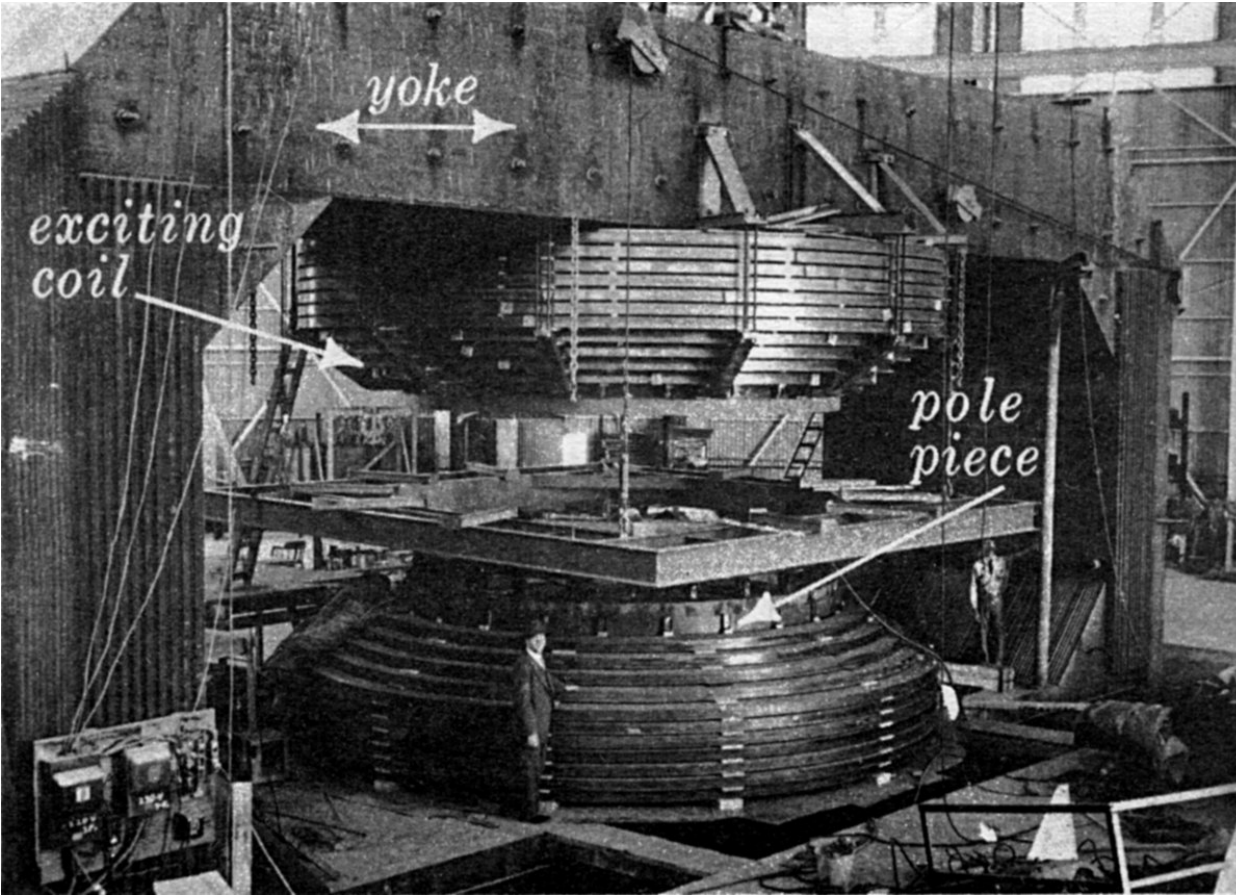
copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

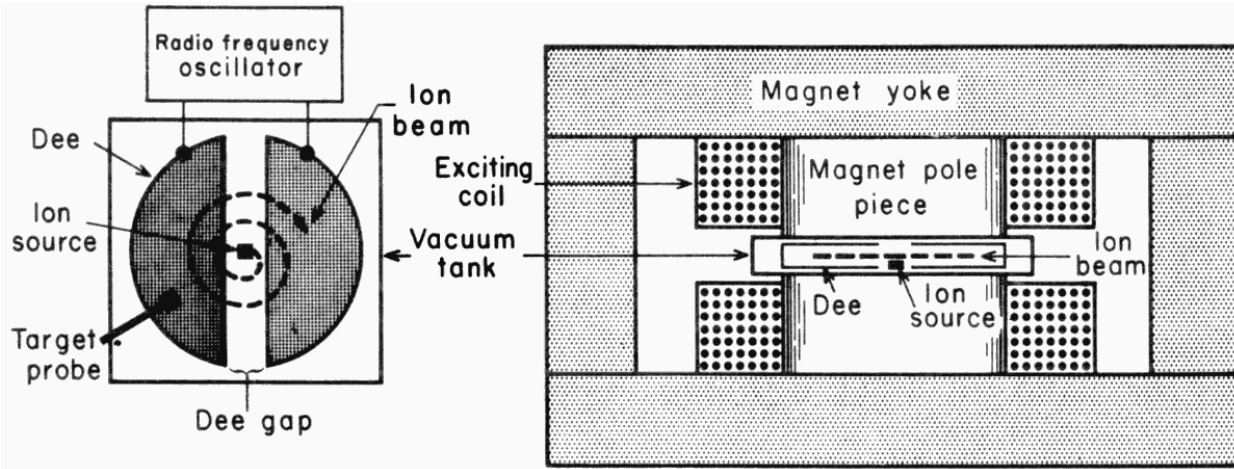
This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.



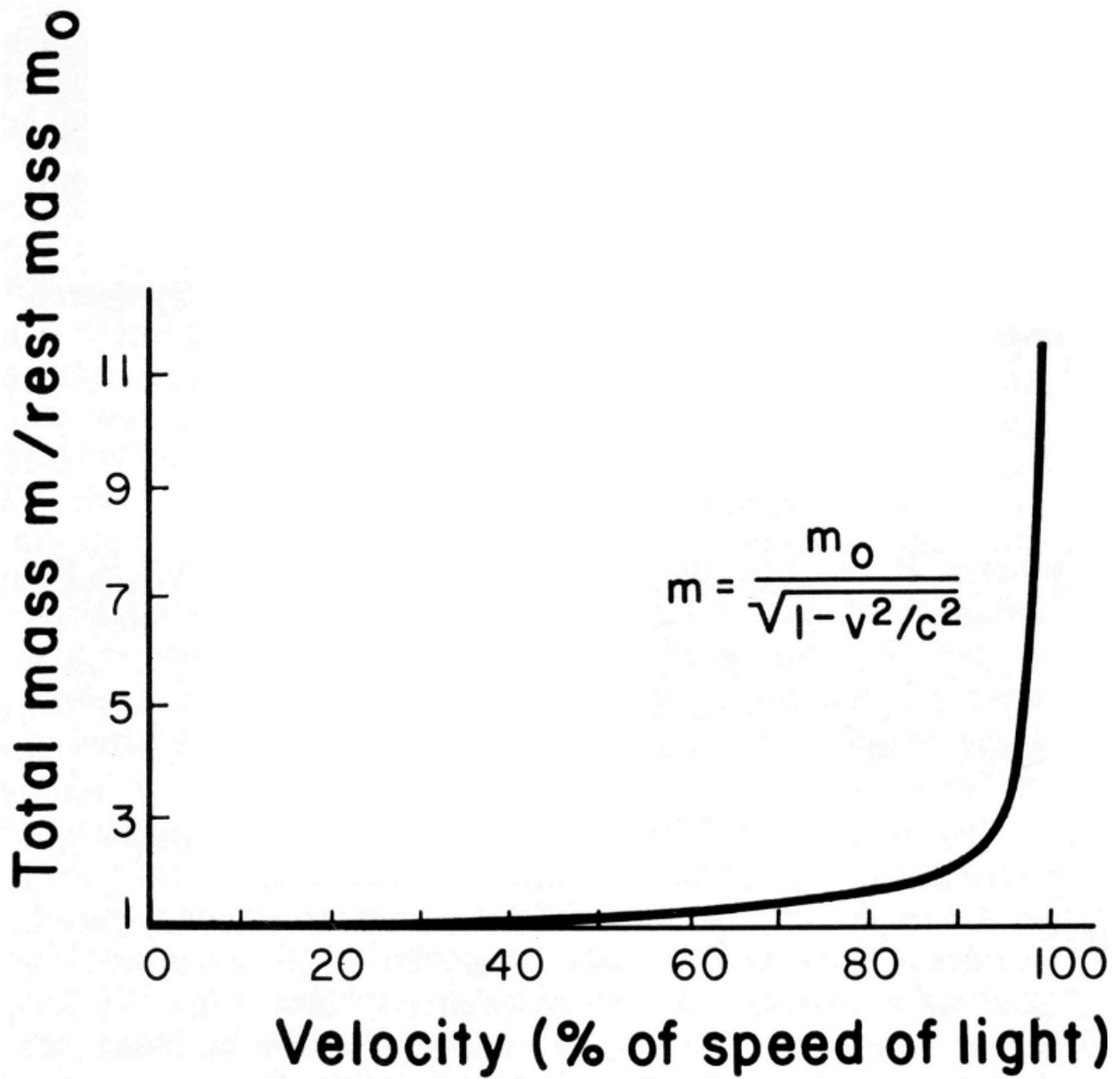
[back](#)



[back](#)

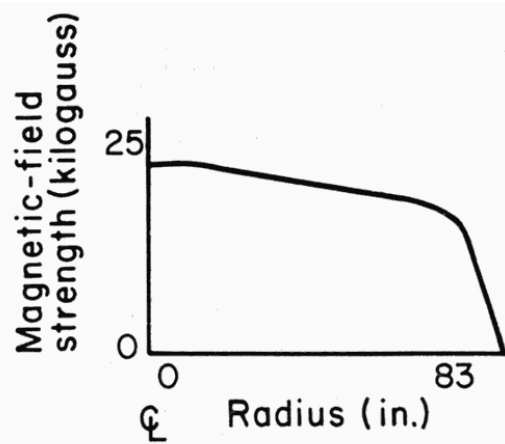


[back](#)

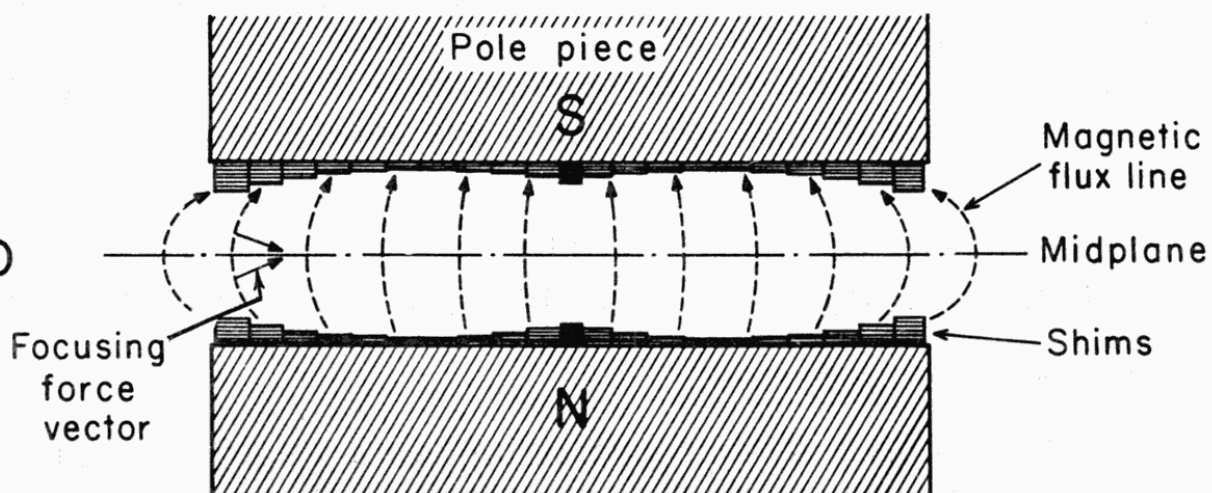


[back](#)

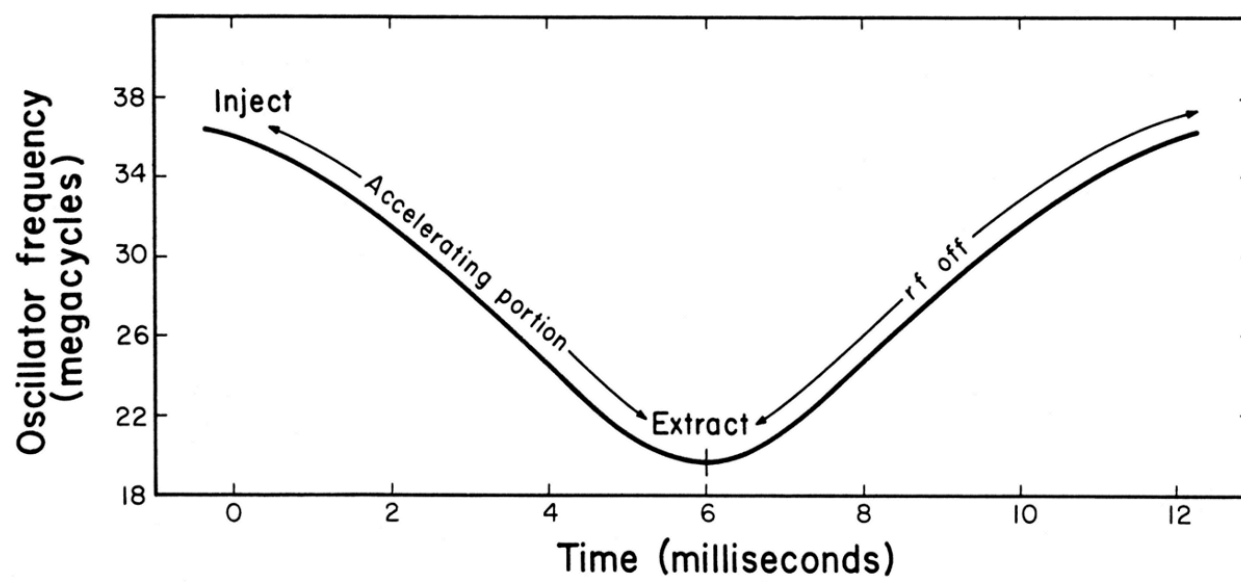
a



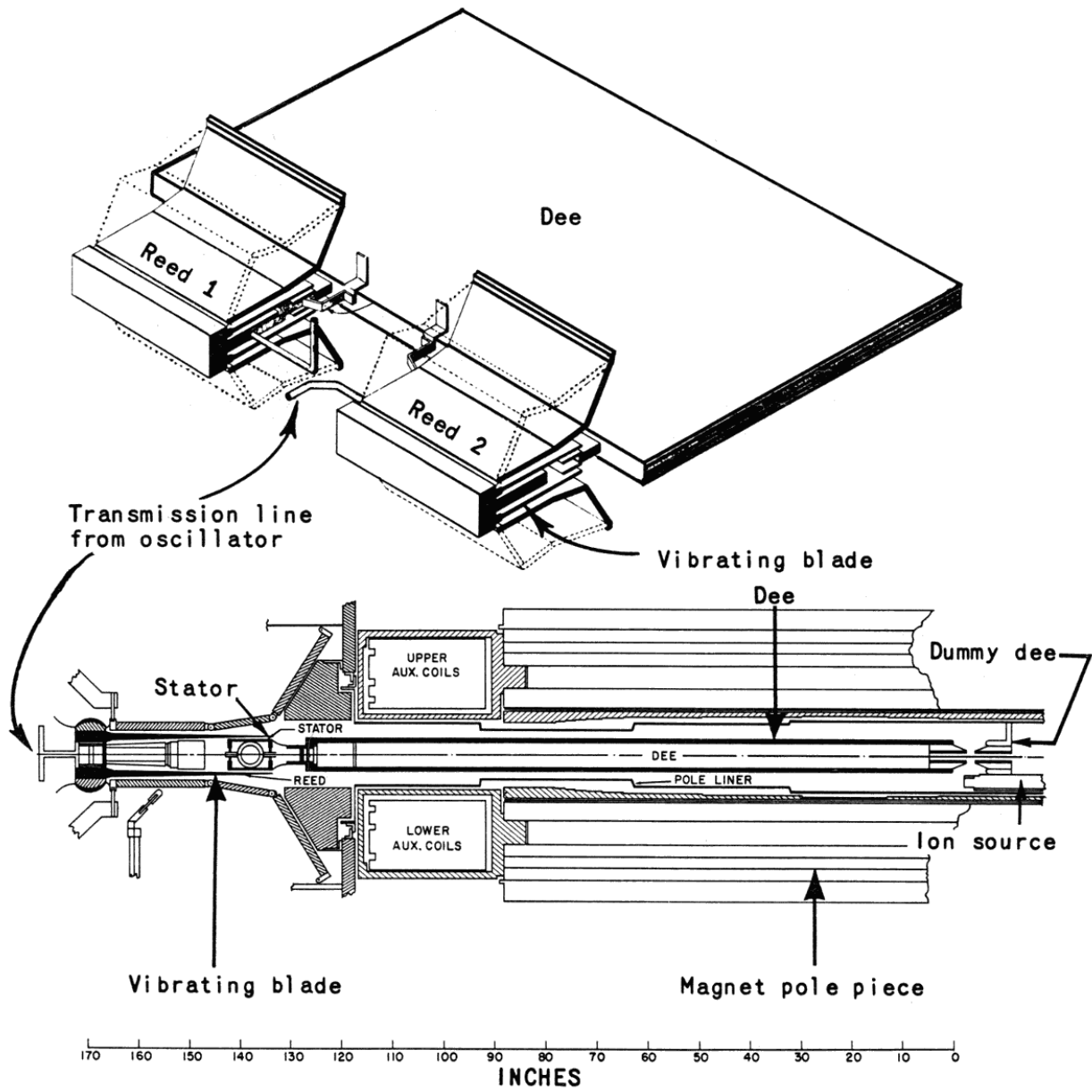
b



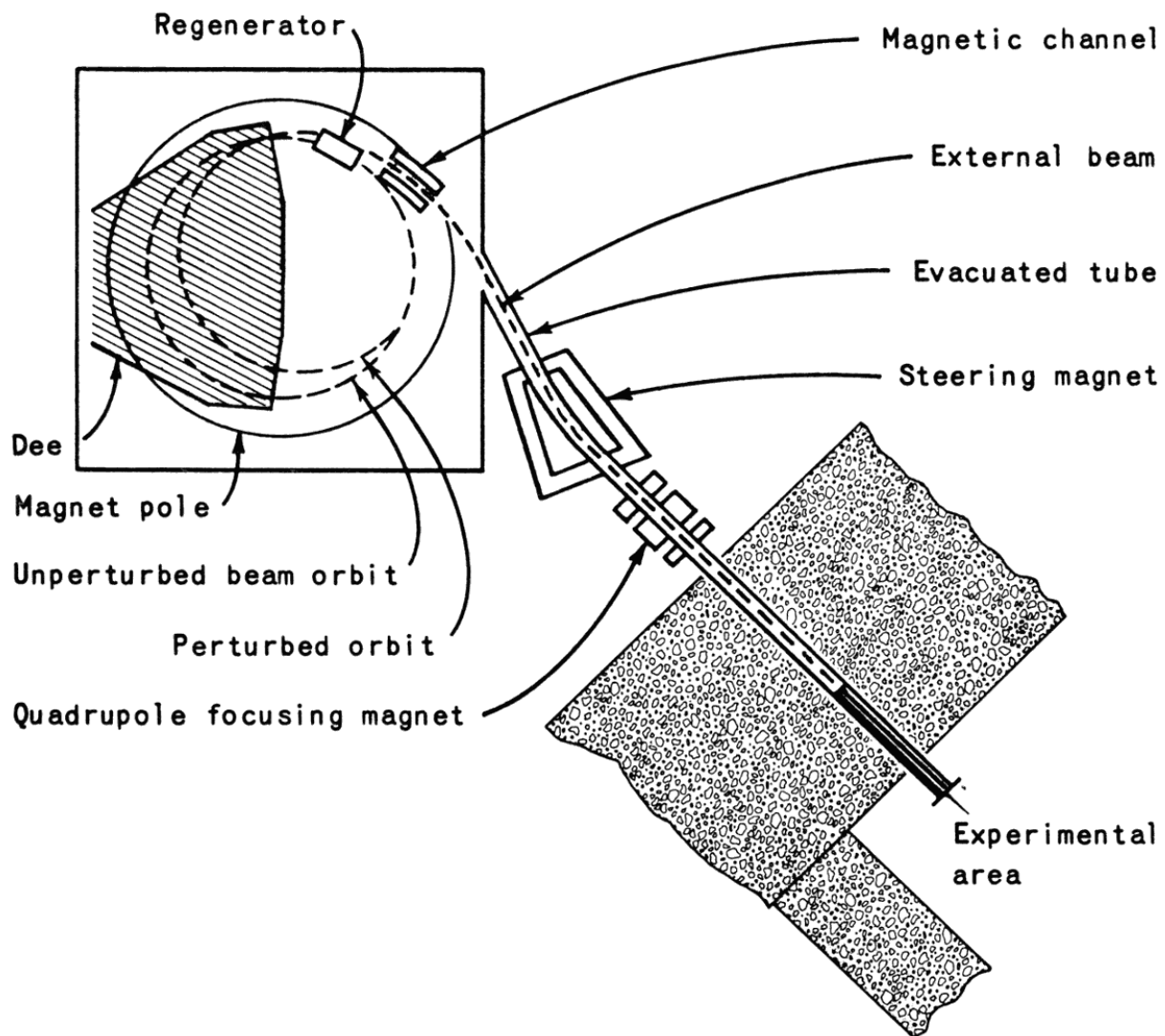
[back](#)



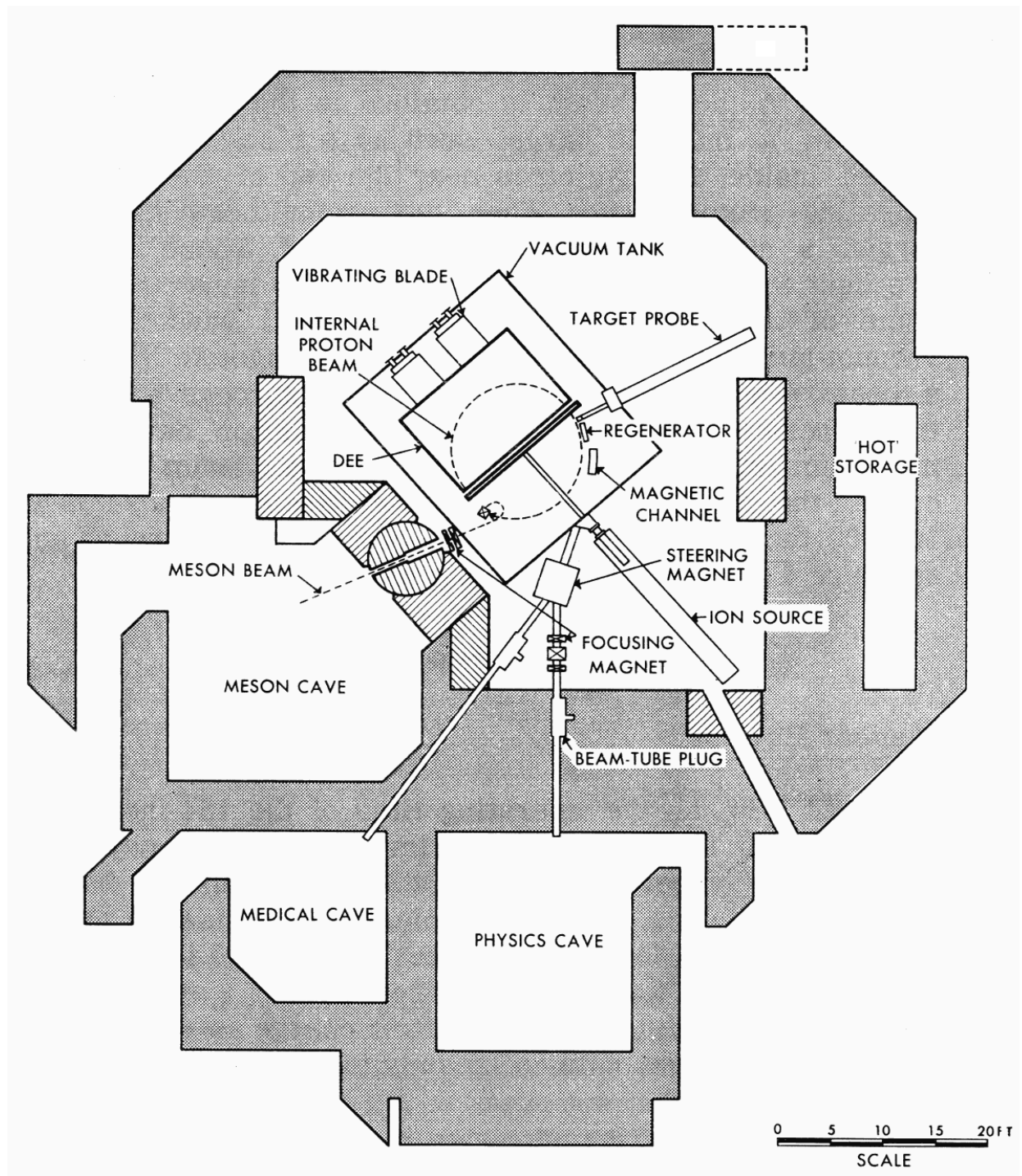
[back](#)



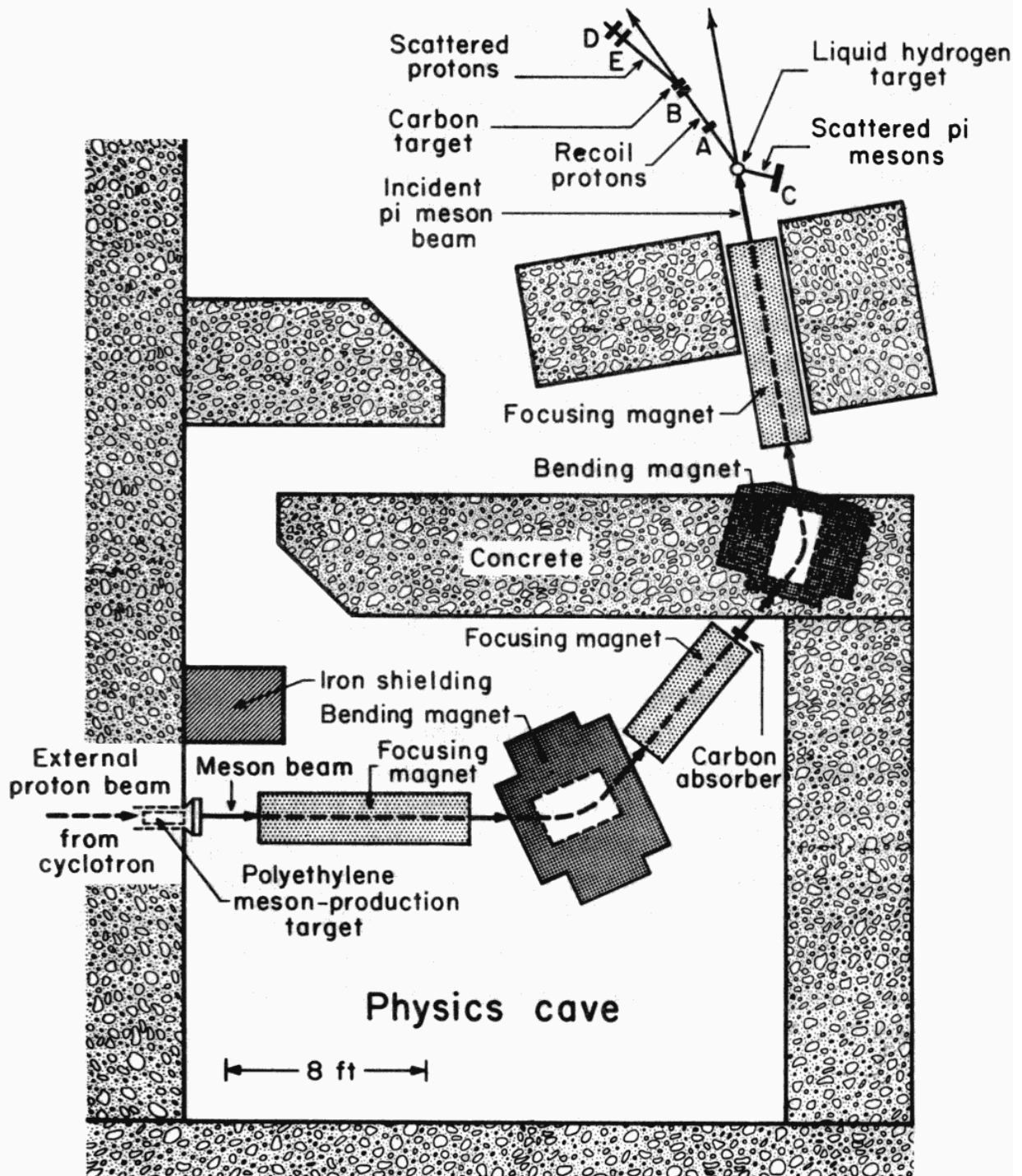
[back](#)



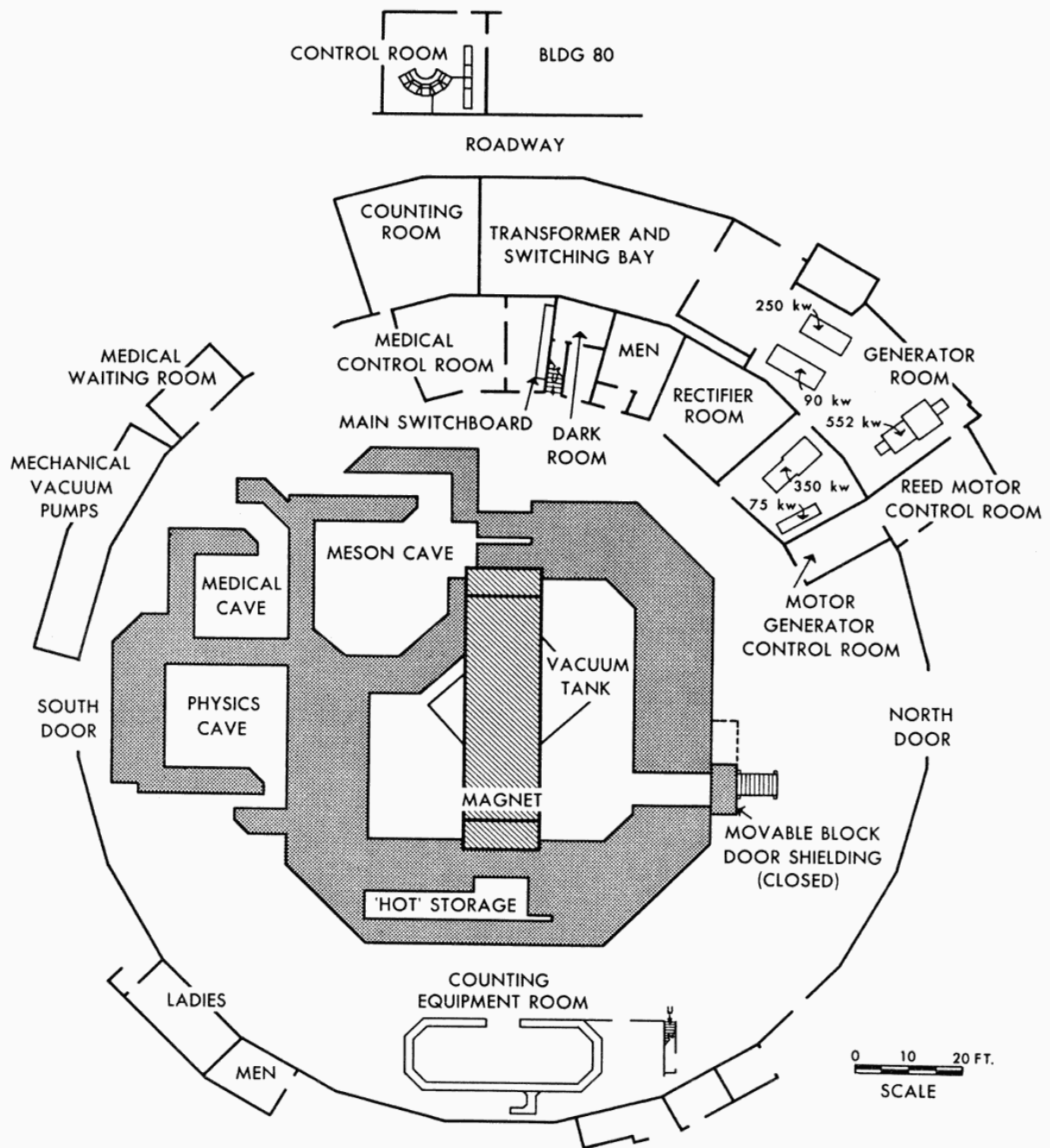
[back](#)



[back](#)



[back](#)



[back](#)